

MULTIPLE DISCRIMINATION BETWEEN DIRICHLET POPULATIONS

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Summary: This paper shows applications of the discriminant function of the Dirichlet distribution in the multivariate and matrix cases. The first is on discriminating between unknown Dirichlet populations with an application on the famous Fisher's Iris data with four variables and three populations. This is a case where the Dirichlet parameters have to be estimated for each population. The second application is on constructing rejection regions for matrix Dirichlet variables with known parameters. The correspondence between the likelihood ratio test statistic for testing equality of covariance matrices and the discriminant function between matrix Dirichlet populations is discussed. The application is on volatility of the Southern Oscillation Index (SOI) for specific months among high, medium and low groupings.

1. Introduction

The Dirichlet distribution is a well-known multivariate distribution (DeGroot, 1970) and is defined as the joint density of $X_i = \frac{Y_i}{\sum_{i=1}^{p+1} Y_i}$, $i = 1, \dots, p+1$ where Y_i is distributed independently $\text{Gamma}(\alpha_i, 1)$.

It is clear that $0 < X_i < 1$ and $\sum_{i=1}^{p+1} X_i = 1$. It is therefore not a $p+1$ dimensional distribution, but a p -variate distribution since $X_{p+1} = 1 - \sum_{i=1}^p X_i$. It is referred to as the composite Dirichlet by Aitchison and Shen (1980) and Aitchison (1986) defined on the simplex $0 < x_i < 1$ and $\sum_{i=1}^{p+1} x_i = 1$ in p dimensions. Aitchison (1986) discussed the advantages of the Dirichlet of which the number of parameters is one more than the number of dimensions. The negative relationship between variables is a big restriction and is overcome by the introduction of the logistic-normal as a much richer class. Compositional data are usually transformed to the Euclidian space where standard statistical methods can be used (see Filzmoser, Hron and Templ, 2012). Due to the need for discriminant analysis in the context of compositional data where the Dirichlet is applicable, we concentrate on the Dirichlet. The

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classification of a Dirichlet observation $\mathbf{x}_0 = (x_{01}, x_{02}, \dots, x_{0p})$ to one of $q > 1$ Dirichlet populations through the discriminant function is considered against the alternative that the observation is just a random observation. An application on Fisher's Iris data is given.

In Section 3 the assignment of a matrix Dirichlet observation to matrix Dirichlet populations is considered. The correspondence between the likelihood ratio test statistic for equivalence of covariance matrices and the discriminant function between two Dirichlet populations is shown. An application is shown on analyzing the volatility of the Southern Oscillation Indexes (SOI) for several months jointly among three groups of indices.

2. The Dirichlet distribution

2.1. Introduction

Discussions on many of the properties of the Dirichlet distribution can be found in Wilks (1962), DeGroot (1970) and Aitchison (1986). We mention a few.

If $X = (X_1, X_2, \dots, X_p)$ is distributed *Dirichlet* $(\alpha_1, \dots, \alpha_{p+1})$ or denoted by $D(\alpha)$ on the positive simplex in p dimensions, then the joint density is given by

$$f(x) = \frac{\prod_{i=1}^{p+1} \Gamma(\alpha_i)}{\Gamma(\alpha_0)} \left\{ \prod_{i=1}^p x_i^{\alpha_i-1} \right\} \left\{ 1 - \sum_{i=1}^p x_i \right\}^{\alpha_{p+1}-1}, \quad 0 < x_i < 1, \sum_{i=1}^p x_i < 1, \alpha_0 = \sum_{i=1}^{p+1} \alpha_i. \quad (1)$$

1. The means and covariances are

$$\begin{aligned} E(X_i) &= \mu_i = \frac{\alpha_i}{\alpha_0}, \quad i = 1, \dots, p \\ \text{Cov}(X_i, X_j) &= \begin{cases} \sigma_{ij} = \frac{-\alpha_i \alpha_j}{\alpha_0^2 (1 + \alpha_0)}, & i \neq j \\ \sigma_{ii} = \frac{\alpha_i (\alpha_0 - \alpha_i)}{\alpha_0^2 (1 + \alpha_0)}, & i = j \end{cases} \end{aligned} \quad (2)$$

$\mu = (\mu_1, \dots, \mu_p)$ denotes the mean of the distribution and $\Sigma = (\sigma_{ij})$ $i, j = 1, \dots, p$ the covariance matrix. Note the negative correlation between variables.

2. Let $Y_i \sim \text{Gamma}(\alpha_i, 1)$, $i = 1, \dots, (p+1)$ independently, then $X_i = Y_i / \sum_{i=1}^{p+1} Y_i$ ($i = 1, \dots, p$) $\sim D(\alpha)$. The joint distribution of $X_i = Y_i / Y_{p+1}$ ($i = 1, \dots, p$) is referred to as the Dirichlet type 2 distribution on the positive p -dimensional space.
3. The marginal distribution of X_i is *Beta* $(\alpha_i, \alpha_0 - \alpha_i)$ and the conditional distribution of X_i given $X_{j \neq i} = x_j, j = 1, \dots, (i), \dots, p$ is a scaled *Beta* (α_i, α_{p+1}) , $0 < x_i < 1 - \sum_{j \neq i}^p x_j$ or $\frac{x_i}{1 - \sum_{j \neq i}^p x_j}$ is distributed *Beta* (α_i, α_{p+1}) .
4. Note that if α is a multiple of β for two Dirichlet distributions $D(\alpha)$ and $D(\beta)$, the means are the same, but the covariance matrices differ.
5. The negative differential entropy $L = E[\log f(X)]$ of the Dirichlet distribution (Equation 1) is given by Honkela (2001) as

$$L = \log \Gamma(\alpha_0) - \sum_{i=1}^{p+1} \log \Gamma(\alpha_i) - \sum_{i=1}^{p+1} (\alpha_i - 1) [\psi(\alpha_0) - \psi(\alpha_i)]. \quad (3)$$

$\psi(\cdot)$ is the digamma function, $\frac{d}{da} \log \Gamma(a) = \psi(a)$.

6. Suppose we have two Dirichlet populations, $D(\alpha_1, \dots, \alpha_p, \alpha_{p+1})$ (parent population) and $D(\beta_1, \dots, \beta_p, \beta_{p+1})$, then the Kullback-Leibler measure of divergence between the two follows from Equation 3 as

$$KL = E \left(\log \frac{f(X)}{f_0(X)} \right) = \log \frac{\Gamma(\alpha_0)}{\Gamma(\beta_0)} - \sum_{i=1}^{p+1} \log \frac{\Gamma(\alpha_i)}{\Gamma(\beta_i)} - \sum_{i=1}^{p+1} (\alpha_i - \beta_i) [\psi(\alpha_0) - \psi(\alpha_i)]. \quad (4)$$

The extent to which $KL \geq 0$ gives a measure of the deviance between f_0 and f .

7. The exponent of L given in Equation 3 is the maximal data information (MDI) prior of the Dirichlet distribution (Equation 1). See Zellner (1997) for information on the MDI prior.
8. Complete compositional independence exists between subsets of X (Aitchison, 1986, 318–319).

2.2. The uniform distribution on p -dimensional simplex

If $\alpha_i = 1$, $i = 1, \dots, p+1$, Equation 1 becomes a uniform distribution. The joint density is given by

$$f(x) = \frac{1}{p!}, \quad 0 < x_i < 1, \quad \sum_{i=1}^p x_i < 1. \quad (5)$$

The distribution is defined on the p -dimensional simplex above.

The mean becomes $\mu = E(X) = \begin{bmatrix} \frac{1}{p+1} \\ \vdots \\ \frac{1}{p+1} \end{bmatrix}$ and the covariance matrix

$$\Sigma = \text{Cov}(X, X') = \frac{1}{(p+1)^2(p+2)} \begin{bmatrix} p & \cdots & -1 \\ \vdots & \ddots & \vdots \\ -1 & \cdots & p \end{bmatrix} \quad \text{with } \Sigma^{-1} = (p+1)(p+2) \begin{bmatrix} 2 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 2 \end{bmatrix}.$$

Let $p = 2$, then given an arbitrarily chosen x_2 , an x_1 can be simulated using $x_1 = (1 - x_2)u_1$ for a random $0 < u_1 < 1$ using the property of the conditional distribution.

The observations are not attracted to the mean at all and the correlation $\rho_{ij} = -\frac{1}{p}$ between any two variables X_i and X_j becomes smaller as the dimension increases.

Visualization of a data set is shown in Figure 1 up to $p = 5$ dimensions.

The point to note is that the observations will be attracted to the mean, irrespective of the number of dimensions, unless the Dirichlet becomes uniform.

2.3. Discriminant function

Fisher's discriminant function in the case of multivariate normal populations is well known (Fisher, 1936; Anderson, 2003; Muirhead, 2005). Bayesian discriminant rules resulting in linear or quadratic group separations based on mainly normal populations also exist (Filzmoser et al., 2012). We consider only discrimination among Dirichlet populations.

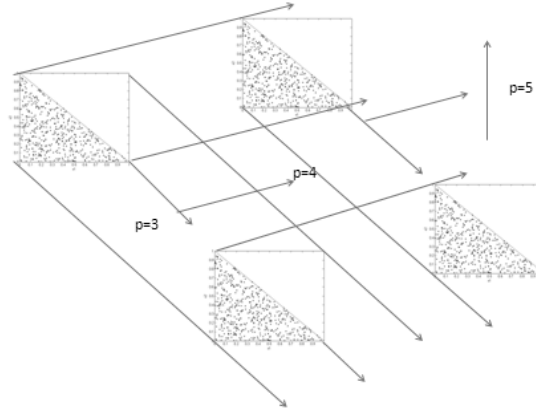


Figure 1: A visualization of the sampled data from the uniform distribution in 2 dimensions to 5 dimensions.

In the case of discriminating between two populations with density functions f_1 and f_2 respectively, an observation $\mathbf{x}_0$ will be assigned to population 1 if $g(\mathbf{x}_0) = \log \left(\frac{f_1(\mathbf{x}_0)}{f_2(\mathbf{x}_0)} \right) > c$ for some constant c . We will consider in Section 2.3.1 the assignment of such an observation if it belongs to one of two Dirichlet populations D_1 or D_2 . In Section 2.3.2 we will extend the analysis to more than two populations. For X_0 random, $E[g(X_0)]$ becomes the Kullback-Leibler measure KL defined in Equation 4 as the divergence measure between f_1 and f_2 .

2.3.1. Two populations

Suppose the random variable $X(p, 1)$ is distributed either $D_1(\alpha)$ or $D_2(\beta)$. If $\mathbf{x}_0(p, 1)$ is observed, then according to the above ratio of the two densities, $\mathbf{x}_0$ will be assigned to D_1 if

$$g(\mathbf{x}_0) = (\alpha - \beta) \log(\mathbf{x}_0) > c. \quad (6)$$

This is a linear function in the elements of $\log(\mathbf{x}_0)$.

The constant c is chosen such that the probabilities of misclassification

$$P[g(X) < c | D_1(\alpha)] \text{ and } P[g(X) > c | D_2(\beta)] \quad (7)$$

are equal.

The distribution of $g(X) = (\alpha - \beta) \log(X)$ is unknown, but can be simulated. The mean is

$$\mu_g = E(g(X) | D_1) = (\alpha - \beta) E[\log(X) | D_1] = \sum_{i=1}^{p+1} (\alpha_i - \beta_i) [\psi(\alpha_i) - \psi(\alpha_0)]. \quad (8)$$

Also $E \log(X_i) = E \log(Y_i) - E \log(\sum_{i=1}^{p+1} Y_i)$ according to the definition of X where Y_i is distributed $Gamma(\alpha_i, 1)$ (see Section 1). It is known that $E \log(Y_i) = \psi(\alpha_i)$. The variance $Var(\log(Y_i)) = \psi'(\alpha_i)$. ψ' is called the trigamma function.

The variance of $g(X)$ is

$$\text{Var}(g(X)|D_1) = (\alpha - \beta)\Phi(\alpha - \beta)', \quad (9)$$

where $\Phi = \text{Cov}(\log X, \log X'|D_1)$.

Since $E[\log(X_i) \log(X_j)] = E\left[\log(Y_i) - \log\left(\sum_{i=1}^{p+1} Y_i\right)\right] \left[\log(Y_j) - \log\left(\sum_{i=1}^{p+1} Y_i\right)\right]$ and $\text{Var}(\log(Y_i)) = \psi'(\alpha_i)$ (Chan, 1993), it follows that the i, j^{th} element of Φ is

$$\phi_{ij} = \text{Cov}(\log X_i, \log X_j|D_1) \approx \begin{cases} (\psi'(\alpha_i) + \psi(\alpha_i)^2 - 2\psi(\alpha_i)\log(\alpha_0) - [\psi(\alpha_i) - \psi(\alpha_0)]^2), & i = j \\ \psi(\alpha_i)\psi(\alpha_j) - \log(\alpha_0)\log(\alpha_i) - \log(\alpha_0)\log(\alpha_j) \\ - [\psi(\alpha_i) - \psi(\alpha_0)][\psi(\alpha_j) - \psi(\alpha_0)], & i \neq j, \end{cases} \quad (10)$$

taking $E\left[\log(Y_i) \log\left(\sum_{i=1}^{p+1} Y_i\right)\right] \approx \log(\alpha_i)\log(\alpha_0)$.

Remark: Note that the joint distribution of $V_i = \frac{X_i}{X_{p+1}}$, $i = 1, \dots, p$ is a Dirichlet type 2 and it follows easily that $\text{Cov}(\log V_i, \log V_j) = \psi'(\alpha_{p+1}) + \psi(\alpha_{p+1})^2 - \psi(\alpha_{p+1})[\psi(\alpha_i) + \psi(\alpha_j)]$.

Example 2.3.1 Consider as an example $n = 30$ observations simulated from two independent Dirichlet populations $D_1(\mathbf{k} = (5, 15, 10))$ and $D_2(\mathbf{l} = (7, 6, 0.6))$. The observations are shown in Figure 3 with dots and their respective means with $\oplus$. Suppose we wish to assign an observation $\mathbf{x}_0 = (0.4, 0.4, 0.2)$ to one of the two populations. According to Equation 7, $c = -20$. The distribution function of $g(X)$ given D_1 and survival function of $g(X)$ given D_2 defined in Equation 7 as simulated are shown in Figure 2. It is clear that we will assign the observation to $D(\mathbf{l})$.

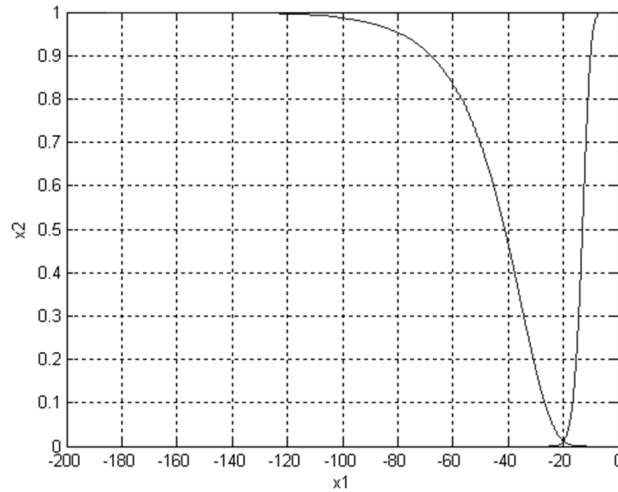


Figure 2: Probabilities of misclassification given $D(\mathbf{k})$ and $D(\mathbf{l})$ respectively. Labels $\mathbf{x}_1$ and $\mathbf{x}_2$ refer to values of $g(X)$ and probabilities respectively.

Suppose the observation $\mathbf{x}_0 = (0.4, 0.1)$ is to be classified, then we rather classify it as a random observation than to assign it to $D(\mathbf{l})$. This is the approach we will follow in the next section when we

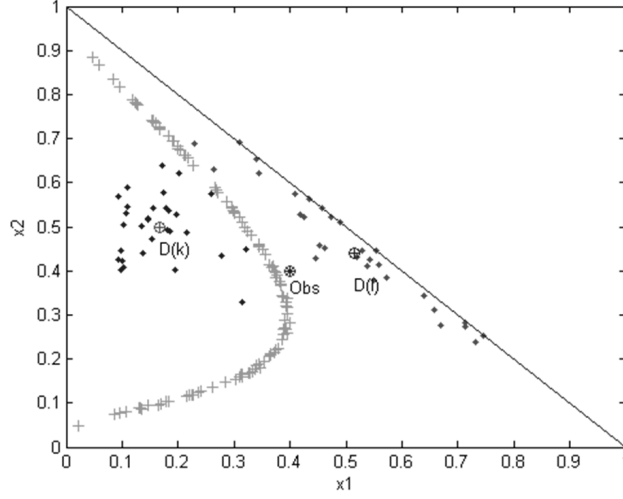


Figure 3: Simulated data from two Dirichlet populations D_1 and D_2 with the discriminant function $g(\mathbf{x}_1, \mathbf{x}_2) = -20$ and the observation $\mathbf{x}_0$ (Obs).

want to assign an observation to one of several populations against the alternative that it is a random observation.

2.3.2. Multiple discrimination

Consider classifying an observation $\mathbf{x}_0 \in D(\alpha^{(i)})$, $i = 1$ or $2, \dots$, or q if

$$(\alpha^{(i)} - \alpha) \log(\mathbf{x}_0) > c_i, \quad i = 1, \dots, q, \quad (11)$$

else consider $\mathbf{x}_0$ as random, namely $\mathbf{x}_0 \in D(\alpha = (1, \dots, 1))$.

Criterion: Select quantiles c_i as critical values such that $P(\text{Type I error}) \approx 0.05$, say,

$$P(X_0 \in D(\alpha) | D(\alpha^{(i)}, i = 1 \text{ or } \dots q)) \approx 0.05. \quad (12)$$

The expectation of the discriminant function for a random $X \in D_j$, against randomness, can be written as

$$\begin{aligned} E[g(X) | D_j] &= (\alpha^{(j)} - \alpha) E \log X \\ &= \sum_{i=1}^{p+1} (\alpha_i^{(j)} - 1) \left[\psi(\alpha_i^{(j)}) - \psi(\alpha_0^{(j)}) \right]. \end{aligned} \quad (13)$$

Consider the following examples:

Example 2.3.2 Consider $q = 4$ Dirichlet populations with $p = 2$ and $\alpha^{(1)} = (10, 20, 3)$; $\alpha^{(2)} = (1, 0.2, 13)$; $\alpha^{(3)} = (16, 2, 3)$; $\alpha^{(4)} = (160, 20, 30)$. Select the quantiles c_i , $i = 1, \dots, 4$ such that the probability of the type I error

$$P(X_0 \in D(\alpha) | D(\alpha^{(i)}, i = 1 \text{ or } \dots q)) \quad (14)$$

is small. A type I error is to assign the observation x_0 as random given it belongs to one of the q populations. Selecting $c = (c_i, i = 1, \dots, 4) = (-30.4437, 1.8616, -160.2341, -14.8681)$, the probability is estimated through simulation, $P(I) = 0.0522$. The discriminating contours

$$(\alpha^{(i)} - \alpha) \log(x) = c_i \quad (15)$$

are obtained through simulating the x -values such that Equation 12 is true (or almost true) for all $i = 1, \dots, 4$. The simulated data is shown in Figure 4, $n = 1000$ for each population, indicated by dark gray (for included observations) and medium gray (for excluded observations) dots. The data points on the right, indicated by dark gray, light gray and medium gray dots, are two overlapping populations, $i = 3, 4$, with the same means but different covariance matrices. An observation $(0.6, 0.1)$ will be assigned to population $i = 3$ while another $(0.7, 0.1)$ to population $i = 4$. An observation $(0.2, 0.1)$ will be considered as random since it cannot be assigned to any of the four populations.

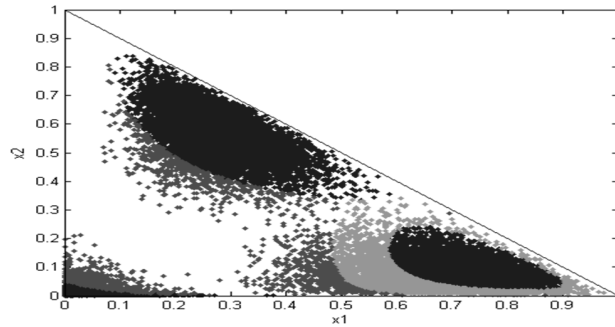


Figure 4: Simulated data from $q = 4$ Dirichlet populations with the discriminant contours.

Example 2.3.3 Fisher's Iris flower data is a well-known dataset with $p = 4$ variables and $q = 3$ populations. It contains observations on the four variables $Y_1 = \text{sepal length}$, $Y_2 = \text{sepal width}$, $Y_3 = \text{petal length}$ and $Y_4 = \text{petal width}$ of Iris-setosa, Iris-versicolor and Iris-virginica respectively measured in *cm*. We will assume that $(Y_1, \dots, Y_4)$ are jointly Dirichlet type 2 distributed (see remark 2 in Section 2.1). Then the distribution of $X = \left(\frac{Y_i}{1 + \sum_{i=1}^4 Y_i}, i = 1, \dots, 4 \right)$ can be described through a Dirichlet distribution $D(\alpha_1, \dots, \alpha_5)$ for each population, say $X^{(i)} \sim D_i \left[\alpha^{(i)} = \left(\alpha_j^{(i)}, j = 1, \dots, 5 \right) \right]$, $i = 1, 2, 3$.

Parameter estimation There are moment, maximum likelihood and Bayes estimates of the parameters $\alpha^{(i)}$ to consider, but we prefer the Bayes estimates since through many simulations they perform the best. The marginal posterior modes from the joint posterior of $\alpha^{(i)}$ were used as the estimates. The maximal data information (MDI) prior of Zellner (see remark 7 of Section 2.1) is used as joint prior of the parameters.

The original data (available in the free package R, 2013, or from Wikipedia at http://en.wikipedia.org/wiki/Iris_flower_data_set) is shown in Figure 5.

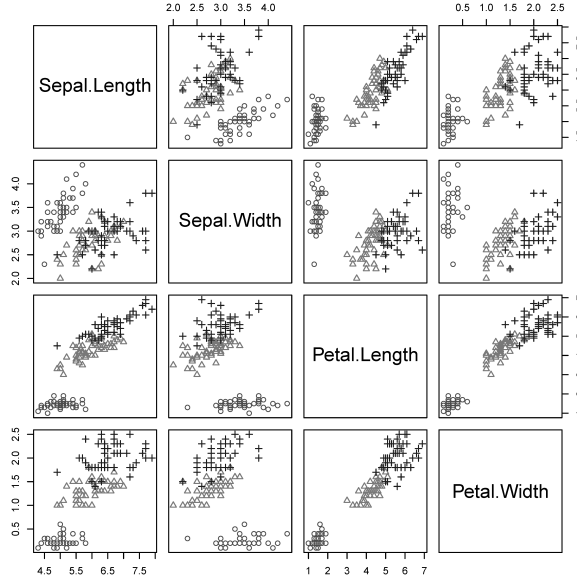


Figure 5: Fisher's Iris flower data. Setosa as circles, Versicolor as triangles and Virginica as plusses.

We proceed to fit Dirichlet distributions on the four variables for each species. We assume each species follows a Dirichlet type 2 distribution and transform the data to Dirichlet type I accordingly. As a measure of goodness of fit, all observations are assigned to its corresponding population under the multiple discriminant rules. From the data, the estimates of the Dirichlet parameters are

$$\begin{aligned}\hat{\alpha}^{(SET)} &= (604.71, 412.57, 175.49, 27.78, 120.02) \\ \hat{\alpha}^{(VER)} &= (378.31, 176.39, 270.87, 84.17, 64.15) \\ \hat{\alpha}^{(VIR)} &= (427.61, 193.26, 360.36, 131.62, 65.37).\end{aligned}\tag{16}$$

A basic test of goodness of fit is to simulate replicate samples and compare them to the original data visually (see for example Gelman, Carlin, Stern, Dunson, Vehtari and Rubin, 2013, 159–171). Doing this with the iris data produces graphs such as Figure 6. There appears to be sufficient correspondence between observed and simulated data.

Figure 6 shows a display of the 50 simulated observations with the 50 observed on each population in 4 dimensions. The fourth variable is indicated by the size of the balls and differences between the populations are obvious. Also the correspondence between the observed and simulated data on the fourth variable is good.

Remark: Instead of assuming a Dirichlet type 2 distribution on Y_i , $i = 1, \dots, 4$, one can think of transforming the data to say Z_i , $i = 1, \dots, 4$ such that it is reasonable to assume that $Z_i \sim \text{Gamma}(\alpha_i, 1)$, $i = 1, \dots, 4$ independently. $X_i = \frac{Z_i}{\sum_{i=1}^4 Y_i}$, $i = 1, \dots, 3$ will then follow a Dirichlet distribution on a 3 dimensional simplex. The advantage is that the number of dimensions is reduced to three.

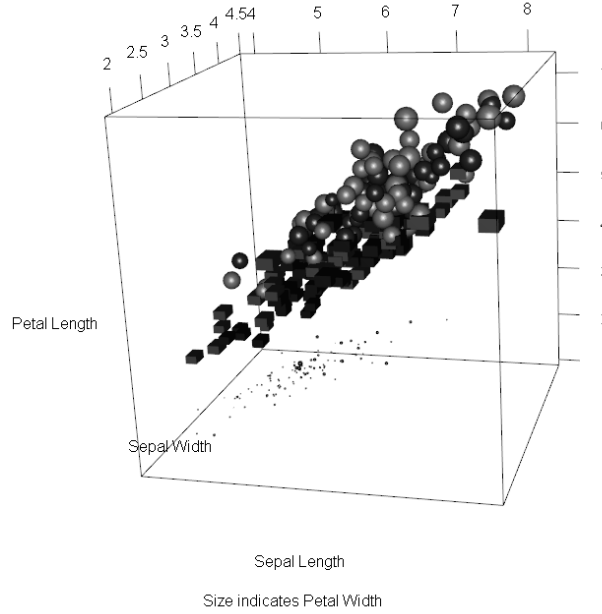


Figure 6: Iris data with simulated replicate samples on Dirichlet Type 2 scale. Dark colours indicate the original data and lighter colours the simulated replicate. Virginica is the top set of spheres, Versicolor is the middle set of cubes and Setosa can be seen at the bottom.

3. The matrix Dirichlet distribution

The matrix Dirichlet distribution came about when Wilks derived the likelihood ratio test statistic for testing the equality of the covariance matrices of multivariate normal populations in 1932 (see also Muirhead, 2005, 291–293). We will consider an application of the discriminant function of this distribution in this section.

3.1. Introduction

Let $A_j \sim \text{Wishart}(I, n_j)$ independent of $B \sim \text{Wishart}(I, m)$.

The joint density of $L_j = \left(\sum_{j=1}^q A_j + B \right)^{-\frac{1}{2}} A_j \left(\sum_{j=1}^q A_j + B \right)^{-\frac{1}{2}}$, $j = 1, \dots, q$ is given by

$$f(L_1, \dots, L_q) = \frac{\Gamma_p(\frac{1}{2}(n+m))}{\prod_{j=1}^q \Gamma_p(\frac{1}{2}n_j) \Gamma_p(\frac{1}{2}m)} \prod_{j=1}^q |L_j|^{\frac{1}{2}(n_j-p-1)} |I - \sum_{j=1}^q L_j|^{\frac{1}{2}(m-p-1)}, \quad 0 < \sum_{j=1}^q L_j < I_p \quad (17)$$

$$\Gamma_p(a) = \pi^{p(p-1)/4} \prod_{i=1}^p \Gamma(a + \frac{1}{2}(1-i)).$$

The expected value of L_j is given by $E(L_j) = \frac{n_j}{(n+m)} I_p$ where $n = \sum_{j=1}^q n_j$.

3.1.1. Matrix uniform

If $n_j = p + 1$ for all j and $m = p + 1$, then Equation 17 becomes the matrix uniform distribution in $q + 1$ dimensional hyperspace with density

$$f(L_1, \dots, L_q) = \frac{\Gamma_p(\frac{1}{2}(q+1)(p+1))}{[\Gamma_p(\frac{1}{2}(p+1))]^{q+1}}, \quad 0 < \sum_{j=1}^q L_j < I_p. \quad (18)$$

The expected value now becomes $E(L_j) = \frac{1}{q+1}I_p$, $j = 1, \dots, q$.

3.1.2. The matrix Dirichlet type 2 distribution

Alternatively, if we define $V_j = B^{-\frac{1}{2}}A_jB^{-\frac{1}{2}}$ where $A_j \sim \text{Wishart}(n_j, I)$ independently of $B \sim \text{Wishart}(m, I)$, the joint density of $V_1, \dots, V_q$ becomes the matrix Dirichlet type 2 density with density function

$$f(V_1, \dots, V_q) = \frac{\Gamma_p(\frac{1}{2}(n+m))}{\prod_{j=1}^q \Gamma_p(\frac{1}{2}n_j) \Gamma_p(\frac{1}{2}m)} \prod_{j=1}^q |V_j|^{\frac{1}{2}(n_j-p-1)} |I + \sum_{j=1}^q V_j|^{-\frac{1}{2}(n+m)}, \quad (19)$$

$n = \sum_{j=1}^q n_j$, $V_j > 0$. Denote it by $M\text{Dirich2}(n_j, j = 1, \dots, q; m)$.

The expected value of V_j is $E(V_j) = \frac{n_j}{m-p-1}I$ which exists for $m > p + 1$. If $p = 1$, then the Dirichlet2 case follows and $\frac{m}{2} > 1$ has been mentioned. This can be derived from the definition $V_j = B^{-\frac{1}{2}}A_jB^{-\frac{1}{2}}$ where $A_j \sim \text{Wishart}(n_j, I)$ independent of $B \sim \text{Wishart}(m, I)$ and using the property $E[E(V_j|B)] = E(V_j)$. Since $B^{-1} \sim \text{InvWishart}(m, I)$, the expected value is $E(B^{-1}) = \frac{1}{m-p-1}I$ and $E(A_j) = n_jI$.

3.2. The discriminant function

The log discrimination function for assigning a matrix Dirichlet observation $L_0 = (L_{0,1}, \dots, L_{0,q+1})$ to one of two populations $MD(n^{(1)}, m^{(1)})$ or $MD(n^{(2)}, m^{(2)})$ is fairly easy to derive by taking the log of the ratio between the two densities at L_0 with parameters $\mathbf{a} = (n^{(1)}, m^{(1)})$ and $\mathbf{b} = (n^{(2)}, m^{(2)})$ as

$$D = (\mathbf{a} - \mathbf{b})[\log |L_{0,1}|, \dots, \log |L_{0,q+1}|]'. \quad (20)$$

The critical value c can be chosen according to Equation 7.

If, however, we would like to assign an observation to a given Dirichlet population or classify it as random, the distribution function of D in Equation 20 can be simulated with $b = (p + 1, \dots, p + 1)$ and c can be calculated from the distribution function. The following example illustrates this.

Example 3.2.1 Let $\mathbf{n} = (15, 13)$ and $m = 5$ be the parameters of a MDir where $p = 3$ and $q = 2$. We want to know if the observation $(|L_{0,1}|, |L_{0,2}|, |L_{0,3}|) = (0.1353, 0.0498, 0.0025)$ can be considered as an observation from the MDir or must be assigned as a random observation. The empirical DF of D with $\mathbf{a} = (15, 13, 5)$ and $\mathbf{b} = (p + 1, p + 1, p + 1)$ from simulated data is shown in Figure 7. The critical 5% value is $c = -71.7046$. The observed value is $d = -55$ which means we consider the observation as belonging to $\text{MDir}(\mathbf{a})$.

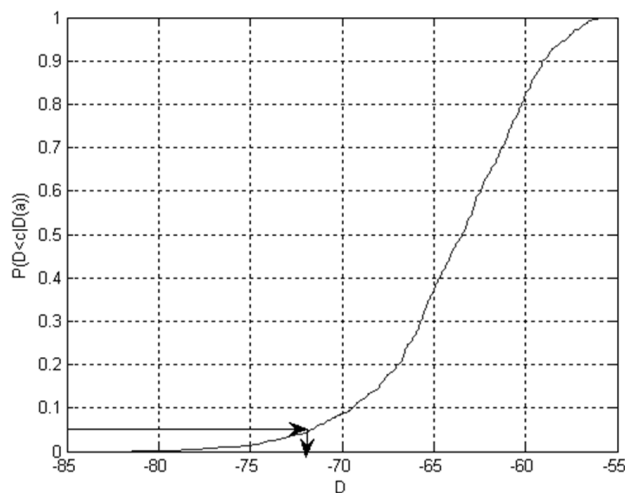


Figure 7: Empirical DF of D . The 5% critical value is $c = -71.7046$.

Example 3.2.2 This is an example where we have $q = 2$ matrix Dirichlet populations $MD_1(15, 3, 5)$ and $MD_2(7, 7, 7)$ with $p = 3$ variables. We need to assign an observation to one of the two populations. The probabilities of misclassification are equal if $c = 23$. Figure 8 shows the DF of the discriminant function D defined in Equation 20 if MD_1 is true and the survival function of D if MD_2 is true from simulated data. An observation with $D = 40$ will be assigned to MD_1 and with $D = 0$ will be assigned to MD_2 .

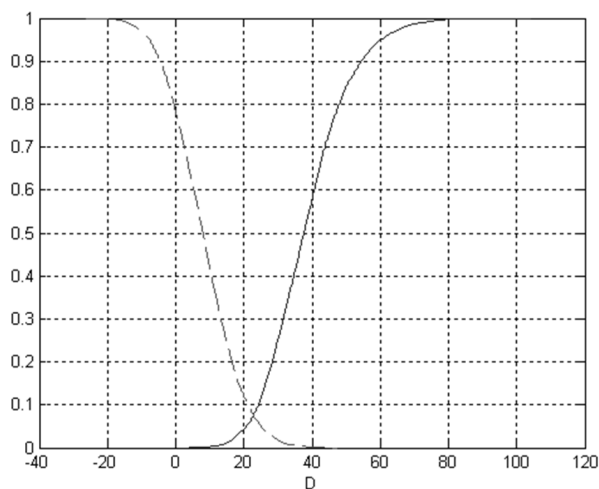


Figure 8: Distribution function of D given MD_1 and survival function (--) of D given MD_2 .

3.2.1. Multiple matrix discrimination

In the case of multiple matrix Dirichlet discrimination, Equation 20 can be applied similar to the procedure described in Section 2.3.2 by taking $b = (p + 1, p + 1, \dots, p + 1)$ and classifying an observation as belonging to one of the populations or randomness.

Example 3.2.3 Consider $q = 3$ matrix Dirichlet populations with $p = 3$, namely $MD_1(15, 8, 5)$, $MD_2(7, 7, 7)$ and $MD_3(7, 9, 15)$. 1000 Matrices were simulated from each population and indicated with dots (dark gray, medium gray and light gray respectively) in Figure 9. From the empirical DF's of D defined in Equation 20 with $\mathbf{a}$ given for each dataset from $MD_i(\mathbf{a})$ above, 53.5% critical values have been calculated, namely $\mathbf{c} = (-46.2085, -35.5028, -62.8077)$ respectively. $\mathbf{b} = (p + 1, p + 1, p + 1)$ in each case. The observations that will be considered as random are encircled in Figure 9. The total number of misclassified observations (out of 3000) are 155, giving an error probability of misclassification of $\frac{155}{3000} = 0.0517$. We can adjust the 53.5% to get a desirable error probability.

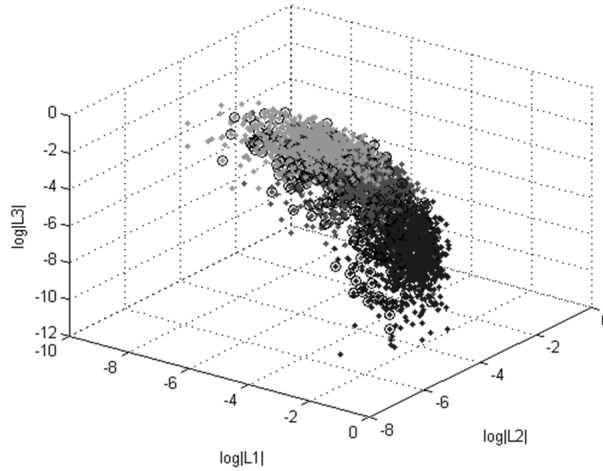


Figure 9: Scatterplot of simulated $\log|L_1|, \log|L_2|, \log|L_3|$ values from each Dirichlet population. Observations encircled are considered as random.

The examples considered in this section are for known parameters of the Dirichlet populations. If the parameters are unknown and the Dirichlet matrices observed, the parameters have to be estimated. This problem has not been considered.

The likelihood ratio test statistic for testing the equivalence between covariance matrices can be written in terms of matrix Dirichlet variables. It is of interest to know what the difference between this statistic and the matrix Dirichlet discriminant function is. We consider this in the next section.

3.3. Testing equality of covariance matrices

Let $L_j = \left(\sum_{j=1}^q A_j \right)^{-\frac{1}{2}} A_j \left(\sum_{j=1}^q A_j \right)^{-\frac{1}{2}}$ where $A_j \sim W(\Sigma_j, n_j)$, $j = 1, \dots, q$ independent of each other. Then the joint distribution of L_j , $j = 1, \dots, q$ is, according to Section 3.1, $Dirichlet(n_j, j =$

$1, \dots, q$) if $\Sigma_j = I_p \forall j$. The log likelihood ratio test statistic for testing the equality of $\Sigma_j = \Sigma$, $j = 1, \dots, q$, ignoring the constant, is given by (Wilks, 1932; Anderson, 2003; Muirhead, 2005)

$$\Lambda = \sum_{j=1}^q n_j \log |L_j|. \quad (21)$$

This statistic is very similar to the discriminant function (Equation 20) if the covariance matrices are equal, and the discriminant function between the *Dirichlet*(n_j , $j = 1, \dots, q$) and randomness, a *Dirichlet*($p + 1, \dots, p + 1$), namely

$$g(L_1, \dots, L_q) = \sum_{j=1}^q (n_j - p - 1) \log |L_j|. \quad (22)$$

The distribution of Λ can be simulated fairly easily and a critical value c can be calculated for small values of Λ below which the assumption of equal covariance matrices will be rejected. The two statistics Equation 21 and Equation 22 are very similar.

The following example gives an illustration:

Example 3.3.1 The southern oscillation index (SOI) is an index based on the air pressure difference between Darwin and Tahiti. This index is used as a leading indicator of rain/drought especially in the southern hemisphere. The monthly SOI values for 1876–2012 are available on the Web. The values are scaled to have mean zero and can be considered to be normally distributed. It has been found that high October SOI values (above 10) indicate good summer rains in the central part of South Africa while a low October SOI (below -10) gives an indication of drought. The summer rain season runs from October to April. We are interested in whether the covariance matrices of the SOI of the months November to April differ if their SOI values are grouped into A , B and C with $A \equiv \text{OctSOI} > 10$, $B \equiv -10 < \text{OctSOI} < 10$ and $C \equiv \text{OctSOI} < -10$. This is considered a case where $p = 6$, $q = 3$ and

$$A_A \sim \text{Wishart}(\Sigma_A, n_A), A_B \sim \text{Wishart}(\Sigma_B, n_B) \text{ and } A_C \sim \text{Wishart}(\Sigma_C, n_C) \quad (23)$$

independent of each other. The degrees of freedom are $n_A = 19$, $n_B = 88$, $n_C = 28$. We wish to know if $\Sigma_A = \Sigma_B = \Sigma_C$. If this is true, $(L_A, L_B, L_C) \sim MD(n_A, n_B, n_C)$ where $L_A = (A_A + A_B + A_C)^{-\frac{1}{2}} A_A (A_A + A_B + A_C)^{-\frac{1}{2}}$. Similarly for L_B and L_C . The empirical distribution function of the simulated values of Λ given in Equation 21 is shown in Figure 10 with the 5% critical value $c = -778.6722$. The observed value is -779.2858 which means we reject the hypothesis of equal covariance matrices and the Dirichlet distribution is invalid. It therefore does not make sense to consider discriminating between $MD(n_A, n_B, n_C)$ and randomness.

4. Conclusion

Discriminating between Dirichlet populations is considered. The extension to matrix Dirichlet populations is shown and the relationship with the test statistic for equality of covariance matrices with the discriminant function illustrated.

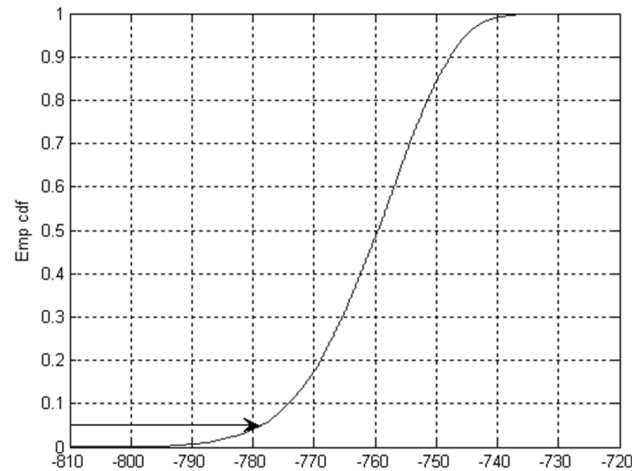


Figure 10: Empirical distribution function of Λ with 5% critical value.

Note: The statistical packages that were used to produce the results illustrated in the figures were MATLAB® (The Mathworks Inc., 2013) and R (R Core Team, 2013). Figure 6 was produced using the RGL package (Adler, Murdoch and others, 2013).

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References

- ADLER, D., MURDOCH, D., AND OTHERS (2013). rgl: 3D visualization device system (OpenGL). R package version 0.93.996.
 URL: <http://CRAN.R-project.org/package=rgl>
- AITCHISON, J. (1986). *The Statistical Analysis of Compositional Data*. Springer Netherlands.
- AITCHISON, J. AND SHEN, S. M. (1980). Logistic-normal distributions: some properties and uses. *Biometrika*, **67**, 261–272.
- ANDERSON, T. W. (2003). *An Introduction to Multivariate Statistical Analysis*. 3rd edition. Wiley.
- CHAN, P. S. (1993). *A Statistical Study of Log-Gamma Distribution*. Open access dissertations and theses, McMaster University.
 URL: <http://digitalcommons.mcmaster.ca/opendissertations/2120>
- DEGROOT, M. H. (1970). *Optimal Statistical Decisions*. McGraw-Hill.
- FILZMOSER, P., HRON, K., AND TEMPL, M. (2012). Discriminant analysis for compositional data and robust parameter estimation. *Computational Statistics*, **27** (4), 585–604. doi:10.1007/s00180-011-0279-8.
 URL: <http://dx.doi.org/10.1007/s00180-011-0279-8>

- FISHER, R. A. (1936). The use of multiple measurements in taxonomic problems. *Annals of Eugenics*, **7** (2), 179–188. doi:10.1111/j.1469-1809.1936.tb02137.x.
URL: <http://dx.doi.org/10.1111/j.1469-1809.1936.tb02137.x>
- GELMAN, A., CARLIN, J. B., STERN, H. S., DUNSON, D. B., VEHTARI, A., AND RUBIN, D. B. (2013). *Bayesian Data Analysis, Third Edition*. Chapman & Hall/CRC Texts in Statistical Science. Taylor & Francis.
- HONKELA, A. (2001). Dirichlet distribution.
- MUIRHEAD, R. J. (2005). *Aspects of Multivariate Statistical Theory*. Wiley.
- R CORE TEAM (2013). R: A language and environment for statistical computing.
URL: <http://www.R-project.org/>
- THE MATHWORKS INC. (2013). Matlab.
URL: <http://www.mathworks.com/>
- WILKS, S. S. (1932). Certain generalizations in the analysis of variance. *Biometrika*, **24** (3-4), 471–494. doi:10.1093/biomet/24.3-4.471.
URL: <http://biomet.oxfordjournals.org/content/24/3-4/471.short>
- WILKS, S. S. (1962). *Mathematical Statistics*. John Wiley & Sons Inc. NY.
- ZELLNER, A. (1997). *Bayesian Analysis in Econometrics and Statistics: The Zellner View and Papers*. Economists of the Twentieth Century Series. Edward Elgar Pub.
URL: <http://books.google.co.za/books?id=ICW7AAAAIAAJ>